

ETSI GS QKD 004 push mode specification, KMS server v1.0.0

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Server description of the KMS endpoint processing ETSI GS QKD 004 push mode requests

Base URLs:

- https://qkd_server/api/v1/qkd/etsi004

Web: Support

etsi004

post__key__material

Code samples

```
# You can also use wget
curl -X POST https://qkd_server/api/v1/qkd/etsi004/key-material \
  -H 'Content-Type: application/json' \
  -H 'Accept: application/json'
```

```
POST https://qkd_server/api/v1/qkd/etsi004/key-material HTTP/1.1
Host: qkd_server
Content-Type: application/json
Accept: application/json
```

```
const inputBody = '{
  "status": null,
  "index": 0,
  "key_buffer": "string",
  "metadata": {
    "size": 57,
    "buffer": {
```

```

        "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"
    }
}
}';
const headers = {
  'Content-Type': 'application/json',
  'Accept': 'application/json'
};

fetch('https://qkd_server/api/v1/qkd/etsi004/key-material',
{
  method: 'POST',
  body: inputBody,
  headers: headers
})
.then(function(res) {
  return res.json();
}).then(function(body) {
  console.log(body);
});

require 'rest-client'
require 'json'

headers = {
  'Content-Type' => 'application/json',
  'Accept' => 'application/json'
}

result = RestClient.post 'https://qkd_server/api/v1/qkd/etsi004/key-material',
  params: {
  }, headers: headers

p JSON.parse(result)

import requests
headers = {
  'Content-Type': 'application/json',
  'Accept': 'application/json'
}

r = requests.post('https://qkd_server/api/v1/qkd/etsi004/key-material', headers = headers)

print(r.json())
<?php

```

```

require 'vendor/autoload.php';

$headers = array(
    'Content-Type' => 'application/json',
    'Accept' => 'application/json',
);

$client = new \GuzzleHttp\Client();

// Define array of request body.
$request_body = array();

try {
    $response = $client->request('POST', 'https://qkd_server/api/v1/qkd/etsi004/key-material',
        array(
            'headers' => $headers,
            'json' => $request_body,
        )
    );
    print_r($response->getBody()->getContents());
}
catch (\GuzzleHttp\Exception\BadResponseException $e) {
    // handle exception or api errors.
    print_r($e->getMessage());
}

// ...

URL obj = new URL("https://qkd_server/api/v1/qkd/etsi004/key-material");
URLConnection con = (URLConnection) obj.openConnection();
con.setRequestMethod("POST");
int responseCode = con.getResponseCode();
BufferedReader in = new BufferedReader(
    new InputStreamReader(con.getInputStream()));
String inputLine;
StringBuffer response = new StringBuffer();
while ((inputLine = in.readLine()) != null) {
    response.append(inputLine);
}
in.close();
System.out.println(response.toString());

package main

import (
    "bytes"
    "net/http"
)

```

```

func main() {

    headers := map[string][]string{
        "Content-Type": []string{"application/json"},
        "Accept": []string{"application/json"},
    }

    data := bytes.NewBuffer([]byte{jsonReq})
    req, err := http.NewRequest("POST", "https://qkd_server/api/v1/qkd/etsi004/key-material", data)
    req.Header = headers

    client := &http.Client{}
    resp, err := client.Do(req)
    // ...
}

```

POST /key-material

endpoint to push key material to.

According to the sequence, after a successful **get_key**, the QKD can push the corresponding key material to this endpoint, whenever it created a new key.

Body parameter

```

{
  "status": null,
  "index": 0,
  "key_buffer": "string",
  "metadata": {
    "size": 57,
    "buffer": {
      "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"
    }
  }
}

```

Parameters

Name	In	Type	Required	Description
body	body	object	true	none
» status	body	any	true	none

Name	In	Type	Required	Description
» index	body	integer(uint32)	true	increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.
» key_buffer	body	string(base64)	true	none
» metadata	body	object	false	none
»» size	body	integer(uint32)	false	size of buffer in characters
»» buffer	body	object(json)	false	metadata in json format.
»»» key_stream_id	body	string(uuid)	true	key stream ID to which the value belongs (required for KMS in case of multiple open key streams)

Example responses

201 Response

```
{
  "status": null,
  "metadata": null
}
```

Responses

Status	Meaning	Description	Schema
201	Created	Key successfully received	Inline

Response Schema

Status Code **201**

Name	Type	Required	Restrictions	Description
» status	any	true	none	none
» metadata	any	false	none	none

This operation does not require authentication